FINC 4101
Investment Analysis

Topic: Bond Pricing and Yields

Learning Objectives

1. Review fixed-coupon bond pricing and the price-yield relationship.
2. Calculate a bond's accrued interest and determine its invoice price.
3. Compute and interpret YTM, HPR, RCY, YTC/YTW, and current yield.
4. Analyze how interest-rate changes, coupon reinvestment, call features, and default risk affect investor returns.
5. Explain how changes in Treasury yields and credit spreads affect bond prices.
6. Select appropriate yield and return measures for different investment horizons and bond features.

The Investor's Question

- A bond is quoted with a price and a yield. What return should the investor actually expect, and what can cause that return to differ from the quoted YTM?
- Can an investor lose money on a Treasury bond even if every promised payment is made?

Fixed-Coupon Bond Cash Flows

- FCB promises a stream of fixed payments (**coupons**) plus a single payment (**face value**) at maturity.
  - Coupon payment = coupon rate × face value ÷ payments per year.
  - Bond price = PV of coupon payments + PV of face value.
  - Unless stated otherwise, assume a $1,000 face value and transactions on coupon dates.

Review: Find a Bond's Price

- An 8% coupon, 30-year bond pays coupons semi-annually. Face value is $1,000, so each six-month coupon is $40.

YTM = 6%

Keys	Input	Output
N	30 x 2 = 60	
I/Y	6/2 = 3	
PV		**- 1,276.76**
PMT	80/2 = 40	
FV	1000	

Premium Bond price = $1,276.76

YTM = 8%

Keys	Input	Output
N	30 x 2 = 60	
I/Y	8/2 = 4	
PV		**- 1,000**
PMT	80/2 = 40	
FV	1000	

Par Bond price = $1000

YTM = 10%

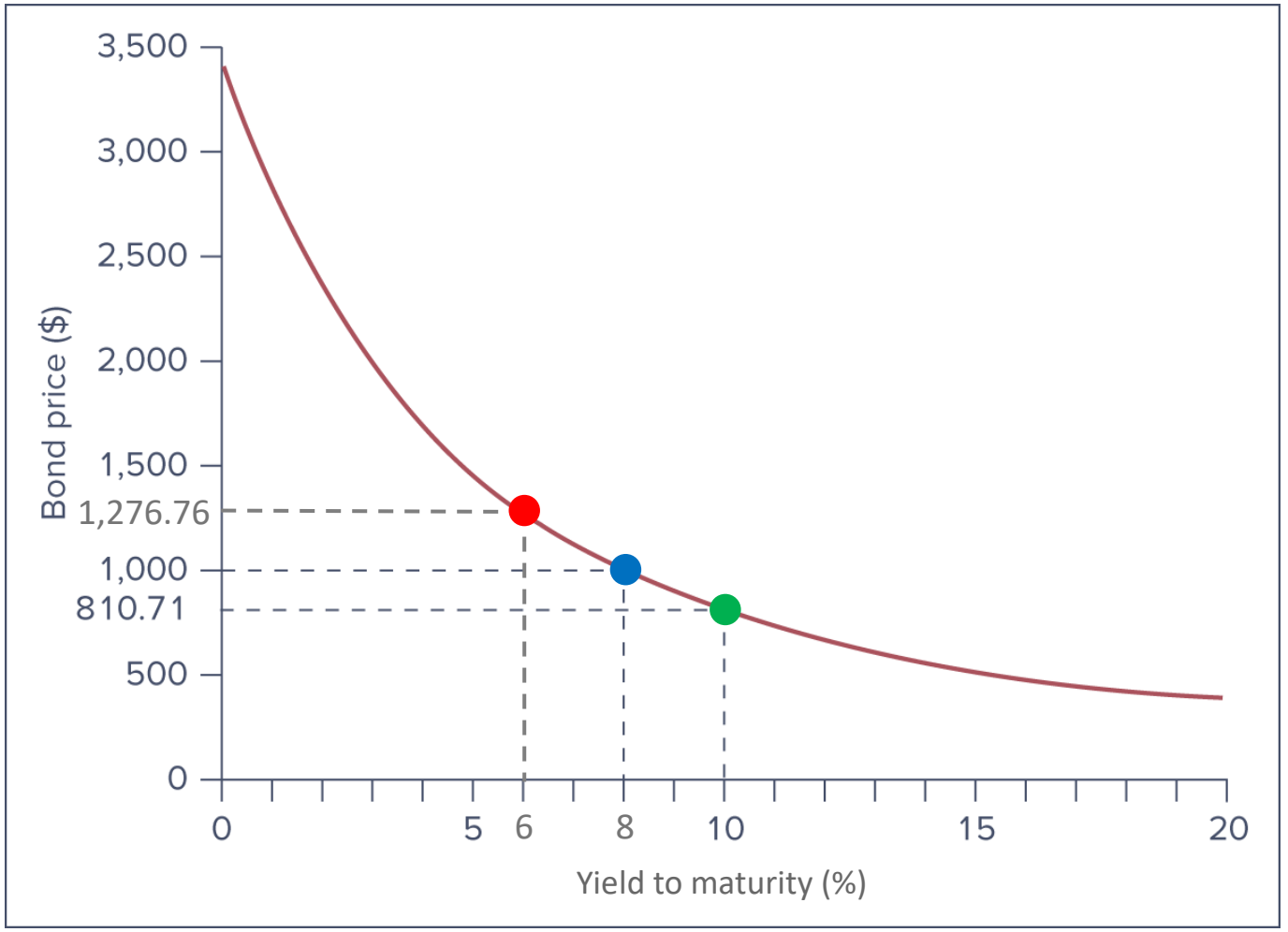
Keys	Input	Output
N	30 x 2 = 60	
I/Y	10/2 = 5	
PV		**- 810.71**
PMT	80/2 = 40	
FV	1000	

Discount Bond price = $810.71

Click here for Bond Valuation Review

The Price–Yield Relationship

- Bond **prices fall** when required **yields rise** and rise when required yields fall.
- **The relationship is curved**: equal yield increases and decreases produce unequal price changes



Bond Price, Coupon Rate & YTM

Relationship	Bond price	Classification
Coupon rate > YTM	Above par	**Premium**
Coupon rate = YTM	At par	**Par**
Coupon rate < YTM	Below par	**Discount**

- A premium or discount reflects how the bond's coupon rate compares with the yield investors require.
- **Does a discount bond necessarily offer a better investment?**

Apply What We Learned

- A bond with an original maturity of 10 years was issued at par four years ago. It pays annual coupons. Its YTM has since fallen from 9% to 7%. Is the bond currently trading:
  A. Below par
  B. At par
  C. Above par
  D. Insufficient information

Invoice Price, Flat Price, Accrued Interest

- **Flat price (clean price):** the quoted bond price, excluding accrued interest.
- **Invoice price (dirty price):** the amount paid by the buyer, including accrued interest.

Invoice price = quoted/flat price + Accrued interest

- For a semi-annual coupon bond:

$$\text{Accrued Interest (A.I.)} = \frac{\text{Annual coupon payment}}{2} \times \frac{\text{Days since last coupon payment}}{\text{Days separating coupon payments}}$$

- Day counts follow the bond's applicable convention.

Accrued Interest Example

- A bond (Par value = $1,000) makes semi-annual payments with a coupon rate of 6%.

1. If 45 days have passed since the last coupon payment, what is the accrued interest, assuming a 180-day semiannual period?
   - $\text{Semiannual coupon} = 0.06 \times 1000/2 = \$30$
   - $\text{Accrued interest} = 30 \times \frac{45}{180} = \$7.5$
2. The quoted price is $990. What is the invoice price?
   - $Invoice\ Price = 990 + 7.5 = \$997.50$

Bond Returns: Which Question Does Each Answer?

Return Measure	Investor Question
Yield to maturity (YTM)	What return is implied if I hold to maturity under standard assumptions?
Holding period return (HPR)	What return did I earn over my actual holding period?
Realized compound yield (RCY)	What compound return do I earn after reinvesting coupons and selling or redeeming the bond?
Yield to Call/Yield to Worst (YTC/YTW)	What yield could I earn if the bond is called, and which redemption scenario gives the lowest yield?
Expected YTM	What yield is implied by the cash flows I expect to receive after allowing for default losses?
Current yield (CY)	How much coupon income do I receive relative to bond price?

Yield to Maturity (YTM)

- YTM is the **internal rate of return** implied by today's price and the bond's promised cash flows.
- It provides a **common comparison** measure across bonds with different coupons and maturities.
- **Quoted YTM is not a guaranteed realized return**.

UNITED STATES TREAS BDS

2.50000% 02/15/2045

Overview | Price & Performance

Basic Analytics

Price (Bid)	71.304
Price (Ask)	71.341
Depth of Book	View
Ask Yield to Worst	4.785%
Ask Yield to Maturity	4.785%
Current Yield	3.504%
Third Party Price	71.215
Spread to Treasuries	0.000
Treasury Benchmark	20 YR.(2.500% 02/15/2045)

Find YTM of a Bond

- Suppose an 8% coupon, 30-year bond is selling at $1,276.76. Coupons are paid semi-annually. What is the bond's **annualized** yield to maturity?
  - Calculator inputs: N = 60; PV = −1,276.76; PMT = 40; FV = 1,000.
  - CPT I/Y = 3% per six months
- **Bond equivalent Yield (BEY)** annualizes the periodic yield *without* compounding

$$BEY = \text{periodic YTM} \times m = 3\% \times 2 = 6\%$$

- **Effective annual yield (EAY)** incorporates compounding

$$EAY = (1 + \text{periodic YTM})^m - 1 = (1.03)^2 - 1 = 6.09\%$$

- BEY and EAY express the **same** periodic yield using different annualization conventions.

Does YTM Equal the Return You Will Actually Earn?

- The investor's compound return matches the initial YTM, on the same compounding basis, if:
- The investor **holds the bond to maturity**.
- All coupons are reinvested at the initial periodic YTM.
- The issuer does not call the bond before maturity.
- The issuer makes every promised payment.

What Happened to Treasury Yields?

- From March 17 to September 17, 2026, the 10-year Treasury yield rose from about 4.20% to 4.94%

- If you bought a 10-year Treasury before this increase and sold it 6 months later, would its sale price be above or below your purchase price? Could the coupon income offset the price change?



Source: CNBC and U.S. Department of the Treasury

What If You Sell Before Maturity?

- Your return now depends on the **price at which you sell the bond**.
- The bond's sale price depends on **market yields at that time**.
- If market yields change, the bond price changes.
- So your realized return can be **higher or lower than the original YTM**.



- How do we measure the return over the period we **actually hold** the bond?

Holding period return (HPR)

- A bond's holding-period return includes coupon income and any price gain or loss over the holding period.

$$HPR = \frac{\text{Price change} + \text{Coupon payment}}{\text{Beginning price}}$$

$$= \frac{\text{Ending price} - \text{Beginning price} + \text{Coupon payment}}{\text{Beginning price}}$$

10-Year Treasury HPR Example

March 17, 2026

- You buy an *illustrative* 10-year Treasury at par.
- Coupon rate = 4.20%
- YTM = 4.20%
- Purchase Price = $1,000
- Semiannual coupon = $21

September 17, 2026

- After receiving one coupon ($21), you sell the bond.
- Remaining maturity = 9.5 years
- Market YTM = 4.94%
- Approximate sale price =$944.43
- $HPR = \frac{944.43-1000+21}{1000} = -3.46\%$

➢ Six-month HPR = −3.46%.

➢ Rising yields caused a **capital loss** that more than offset the coupon income.

Based on 10-year Treasury yield movements, Mar. 17 – Sept. 17, 2026.

What If the Holding Period Is Longer?

- Over several years, the investor may receive **multiple coupon payments**.
- Those coupons can be **reinvested at rates different from the original YTM**.
- The bond may also be sold at a price different from its purchase price.
- We therefore need a return measure that incorporates **both reinvestment income and the terminal sale price**.

➢ This is the idea behind **Realized Compound Yield (RCY).**

Realized Compound Yield

- Realized Compound Yield is the compound rate of return over the investor's holding period.
- **RCY incorporates**:
  1. Reinvested coupon payments, and
  2. The bond's sale price or redemption value at the end of the holding period.

Realized Compound Yield Calculation

Length of holding period

$$y = \left(\frac{Terminal\ wealth}{Purchase\ price}\right)^{\frac{1}{n}} - 1$$

- **Terminal wealth** includes the sale price or redemption value **plus** the future value of coupon payments
- n = length of holding period (could be in years, half-years, quarters, etc.).
- The calculated **yield is per period**: if n is in years, the yield is annual; if n is in half-years, the yield is semiannual.

Realized Compound Yield Example

- Suppose you bought a 30-year, 7.5%, **annual** payment coupon bond for $980 and plan to hold it for 20 years.
- You estimate the bond's YTM will be 8% when it is sold and that the reinvestment rate on the coupons will be 6%.
- Assume the bond is sold immediately after receiving the twentieth coupon.
- Compute the annual RCY under these assumptions.

Realized Compound Yield Solution

Inputs: P_0=$980, FV=$1,000, coupon = 7.5% annual ($75), maturity = 30 yrs, holding period = 20 yrs, remaining maturity = 10 yrs, sale YTM = 8%, reinvestment rate = 6%.

1. Forecast the sales price with 10 years until maturity.
2. Find the FV of 20 coupons reinvested at 6%.

Keys	Input	Output
N	10	
I/Y	8	
PV		**- 966.45**
PMT	75	
FV	1000	

Keys	Input	Output
N	20	
I/Y	6	
PV	0	
PMT	75	
FV		**- 2758.92**

Realized Compound Yield Solution Continued

3. Terminal wealth

   $= \$966.45 + \$2758.92 = \$3725.37$

4. Realized compound yield

$$= \left(\frac{3725.37}{980}\right)^{\frac{1}{20}} - 1 = 6.90\%$$

➢ Annual realized compound yield = 6.90%

- Purchase price = $980
- Terminal wealth = $3,725.37
- Holding period = 20 years

Keys	Input	Output
N	20	
I/Y		**6.90**
PV	-980	
PMT	0	
FV	3725.37	

Yield to Maturity vs. Realized Compound Yield

- For a coupon bond held to maturity, with all promised payments made, compare RCY and YTM on the same compounding basis:
  - RCY < YTM, if coupon reinvestment rate < YTM
  - RCY = YTM, if coupon reinvestment rate = YTM
  - RCY > YTM, if coupon reinvestment rate > YTM
- YTM is **NOT** a guaranteed realized return because future reinvestment rates are uncertain.

What If the Bond Is Called Early?

- A callable bond gives the **issuer** the right to repay the bond ***before*** maturity at a price specified in the bond's call terms
  - This is especially attractive to the issuer when market rates fall.
- The investor may receive the **call price** instead of holding the bond to maturity.
- Investors should also consider the yield implied by an early call.

➢ This leads to **Yield to Call** (YTC): the yield implied by today's price if the bond is called on a specified date at the specified call price.

Callable Bond Example: Devon Energy as of 02/23/2026

Call Schedule

DEVON ENERGY CORP NEW SER B NOTE 4.50000%	01/15/2030
Call Defeased	NO
Continuously Callable	YES
Callable After	01/15/2025
Next Call Date *	03/06/2026
Extraordinary Redemption	NO
Make Whole Call	NO
Conditional Call	NO

* The Next Call Date is, where applicable, the earlier of either the continuously callable effective date calculated from the current business day, or the next scheduled call date.

Call Date	Call Price	Call Type
01/15/2025	102.250	Premium
01/15/2026	101.500	Premium
01/15/2027	100.750	Premium
01/15/2028	100.000	Par Call

DEVON ENERGY CORP NEW SE

4.50000% 01/15/2030

Overview | Price & Performance

Basic Analytics

Price (Bid)	100.795
Price (Ask)	100.990
Depth of Book	View
Ask Yield to Worst	3.950%
Ask Yield to Maturity	4.220%
Current Yield	4.455%

Why Does Callability Matter to Investors?

- When rates fall, the bond's price would normally rise.
- But the issuer is more likely to call the bond.
- **If the bond is called**, the investor **loses** the remaining high coupons and faces **lower** available rates when reinvesting the proceeds.
- **Callability therefore limits some of the investor's upside when rates fall.**

Yield To Call Example

- A 20-year maturity bond with par value $1000 makes semi-annual coupon payments at a coupon rate of 8%.
- The bond is currently selling for $1,150 and is ***callable*** in **10** years at a call price of $1,050.
- If the bond is called in 10 years, what return is implied by today's price?

Keys	Input	Output
N	**10 x 2 = 20**	
I/Y		**3.16**
PV	-1150	
PMT	40	
FV	**1050**	

Annualized YTC = 3.16% x 2 = 6.32%

For comparison, YTM = 6.63%

YTM vs. YTC: Which One Matters?

- **YTM = 6.63%:** assumes the bond remains outstanding until maturity.
- **YTC = 6.32%:** assumes the bond is called on the specified call date.
- Why should the investor consider YTC?
  - An early call would produce a lower yield than holding the bond to maturity.

➢ **Yield to Worst (YTW)**: the lowest yield among the relevant redemption scenarios, assuming no default.

What If the Issuer Cannot Make the Promised Payments?

- YTM is calculated using the bond's **promised coupon and principal payments**.
- If **default** is possible, the cash flows investors actually receive may be lower.
- Investors therefore demand compensation for **credit/default risk**.
- **A high promised YTM does not necessarily mean a high expected return.**

Promised Yield vs. Expected YTM

- **Promised YTM**
  - Calculated using all **contractual cash flows**.
  - Assumes the issuer makes every promised payment.
- **Expected YTM**
  - Calculated using **expected** cash flows rather than fully promised payments.
  - Incorporates possible default losses or reduced recovery.
- Credit risk creates a gap between the return promised by the bond and the return investors expect to earn.

Promised vs. Expected YTM Example

- **Bond terms**
  - Coupon rate = **9%**
  - Coupons = **semiannual**
  - Remaining maturity = **10 years**
  - Current price = **$750**
  - Face value = **$1,000**

- **Investor expectation**
  - Assume all remaining coupons are paid, but **only $700 of principal** is repaid at maturity.

1. **What is the promised YTM?**
2. **What yield is implied by the expected cash flows?**

Promised vs. Expected YTM Example

	Promised YTM	Expected YTM
Principal at maturity	$1,000	$700
Annualized yield	13.7%	11.6%

- The promised YTM, which is based on promised payments, is 13.7%.
- Based on the expected payment of $700 at maturity, the yield is only 11.6%.
- Under the assumed cash flows, the implied yield falls from 13.7% to 11.6%.

Credit Ratings and Default Risk

	Bond Ratings			
	Very High Quality	High Quality	Speculative	Very Poor
Standard & Poor's	AAA AA	A BBB	BB B	CCC D
Moody's	Aaa Aa	A Baa	Ba B	Caa C

Investment grade
Relatively lower assessed credit risk

Speculative grade
Relatively higher assessed credit risk

- Why would a lower-rated bond generally need to offer a higher yield?

Credit Spreads and Credit Risk

- Investors require additional compensation for bearing **credit risk**.
- Corporate bonds therefore generally offer higher yields than Treasury securities with **similar maturities**.
- The difference is called the **credit spread**.

$$\text{Credit Spread} = YTM_{\text{Corporate Bond}} - YTM_{\text{Treasury}}$$

- Higher perceived credit risk → wider credit spread

Credit Spread Example

- Corporate bond YTM = **6.8%**
- Comparable Treasury YTM = **4.3%**

$$\text{Credit Spread} = 6.8\% - 4.3\% = 2.5\% = 250 \text{ bps}$$

- Higher perceived credit risk → wider spread → higher required yield → lower bond price, all else equal.

Treasury Yields and Credit Spreads

- A corporate bond initially has:
  - Treasury benchmark yield = 4.3%
  - Credit spread = 250 basis points
  - Corporate bond yield = 6.8%
- Suppose the **Treasury yield falls** by 50 basis points, while the credit spread widens by 100 basis points.
- Questions:

1. What is the new corporate bond yield?
2. Does its price rise or fall, all else equal?

Current Yield

$$CY = \frac{\text{Annual Coupon}}{\text{Bond Price}}$$

- Measures **annual** coupon income relative to current price.
- Ignores capital gains or losses.
- Ignores coupon reinvestment income.
- **Coupon Rate, Current Yield, and YTM**

Premium bond: Coupon Rate $> CY > YTM$

Par bond: Coupon Rate $= CY = YTM$

Discount bond: Coupon Rate $< CY < YTM$

Key Takeaways

- **YTM:** What return is implied if everything follows the maturity scenario?
  - **Sell before maturity**
    Sale price changes with yields → **HPR**
  - **Multiple coupons and reinvestment**
    Terminal wealth depends on sale price + reinvested coupons → **RCY**
  - **Issuer may call the bond early**
    Call price + call date → **YTC / YTW**
  - **Issuer may not make promised payments**
    Default risk → Promised YTM vs. Expected YTM
- The quoted YTM is a useful benchmark, but not necessarily the return the investor will actually earn.

What Causes a Bond's Price to Change?

1. **Passage of time**
   - If YTM stays unchanged and promised payments are made, premium and discount bonds move toward par as maturity approaches.
2. **Changes in market yields**
   - Market yields rise → bond prices fall.
   - Market yields fall → bond prices rise.
   - Interest-rate changes create price risk, especially for investors who may sell before maturity.

- Do all bonds respond equally to the same change in market yields?
  - **Duration** helps us measure this sensitivity.

Practice Questions for Exam 2

- Chapter 10 problem set 1:
  - 4. A coupon bond paying…
  - 6. Why do bond prices…
  - 7. Two bonds have identical times…
  - 8. Consider a bond with a 10% …
  - 12. You buy an eight- year maturity…
  - 14. Which security has a higher…
  - 17. A 20- year maturity bond…
  - 22. A bond with a coupon rate of 7%...
  - 29. Suppose that today ’s date is …
- Chapter 10 problem set 2:
  - 9. A bond with an annual coupon…
  - 16. Consider a bond paying a coupon…
  - 23. A bond has a current yield of 9%...
  - 24. Is the coupon rate of the bond…
  - 28. A two-year bond with par value…
  - 32. A 30-year maturity , 6% coupon…

Additional Practice & Reference

Additional Practice: Realized Compound Yield

Five years ago, XYZ Inc issued a 5% **semi-annual** payment coupon bond with a maturity of ten years. You buy the bond now at a price of $983.94 and plan to hold it for three years. You forecast that you can invest the coupon payments at a stated annual rate of 6.25% and that at the end of three years, the yield will be 7.75%.

- Compute the effective annual RCY.

Additional Practice: Solutions

- Bond price at the end of 3 years, $949.9416
- Find the future value of coupons, $162.22
- Total proceeds = 949.9416 + 162.2186 = 1112.16
- **Effective annual** realized compound yield:

  = $(1112.16/983.94)^{1/3} - 1 = 4.17\ \%$

Keys	Input	Output
N	4	
I/Y	3.875	
PV		**- 949.94**
PMT	25	
FV	1000	

Keys	Input	Output
N	6	
I/Y	3.125	
PV	0	
PMT	25	
FV		**- 162.22**

Keys	Input	Output
N	3	
I/Y		**4.17**
PV	-983.94	
PMT	0	
FV	1112.16	

Promised vs. Expected YTM Example Detailed Solutions

- For promised YTM, FV is 1000,
- For expected YTM, FV is 0.7 x 1000 = 700.

Promised / Stated YTM:

I/Y = 6.8285% (semi-annual rate).

The annual YTM = 6.8285 x 2 = 13.657 or 13.7% p.a.

Expected YTM:

I/Y = 5.8151% (semi-annual rate).

The annual YTM = 5.8151 x 2 = 11.6302 or 11.6% p.a.

Keys	Input	Output
N	20	
I/Y		**6.8285**
PV	-750	
PMT	45	
FV	1000	

Keys	Input	Output
N	20	
I/Y		**5.8151**
PV	-750	
PMT	45	
FV	700	